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**Booms, Busts, and Mismatch in Capital Markets: Evidence from the Offshore
Oil and Gas Industry**

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1 Big picture

- **Question.** How efficiently do decentralized capital markets reallocate capital during booms and busts?
- **Setting.** The paper studies shallow-water offshore drilling in the U.S. Gulf of Mexico from 2000–2009. Oil and gas companies own leases and wells, but they typically do not own drilling rigs. They must search for rigs in a decentralized capital market.
- **Objects being matched.** Projects are wells, ranked by an engineering measure of well complexity. Capital is drilling rigs, ranked by maximum water depth and grouped into low-, mid-, and high-efficiency rigs.
- **Main mechanism.** Booms increase the option value of searching. When future matching opportunities are better, rig owners and well owners become more selective and avoid poor matches. This creates a **sorting effect**: high-efficiency rigs are more likely to drill complex wells, while low-efficiency rigs focus on simpler wells.
- **Why this is not mechanical.** Higher oil and gas prices also raise the current value of any match. If the match-value effect dominated, agents might accept more matches and sorting could weaken in booms. The empirical question is whether the option-value effect dominates the match-value effect.
- **Empirical facts.** The data show positive assortative matching, strong boom-bust cycles in rig prices, and stronger sorting in booms than in busts.
- **Identification challenge.** Observed boom-time matches can change either because agents sort better or because the composition of searching wells changes. The model separates these channels using prices, search behavior, and a structural demand process.
- **Structural model.** The model combines sequential search, two-sided vertical heterogeneity, partially directed search, endogenous entry, Nash bargaining, and dynamic rig-owner outside options.
- **Main counterfactual: sorting.** Removing the sorting effect lowers welfare. Moving from the no-sorting counterfactual to the market allocation increases welfare by about 12%.
- **Market-design counterfactual.** A strong intermediary that reduces search frictions increases welfare by about 51%, mostly by improving match quality rather than simply increasing the number of matches.
- **Policy counterfactual.** Demand smoothing changes drilling activity substantially, but the overall welfare gain is modest, about 14.9%, and it can reduce match quality in booms.
- **Core takeaway.** In this capital market, booms are not only periods of more activity. They are periods in which the market reallocates capital more selectively, and that selectivity is economically valuable.

2 Data and measurement

2.1 Industry setting

- The industry is shallow-water drilling in the U.S. Gulf of Mexico, before the post-Deepwater Horizon regulatory shift.
- Lease holders such as large oil companies choose projects but typically procure drilling rigs from independent rig owners.
- The market is decentralized: firms negotiate contracts bilaterally, sometimes with brokers, and the process is not a centralized auction or exchange.
- Contracts are fixed-price day-rate contracts for a fixed duration, and many are extended to drill another well.
- The author focuses on natural gas prices because shallow-water Gulf of Mexico wells in the sample contain substantial natural gas, and gas prices drive drilling demand.

Interpretation: The market is useful because it is cyclical, decentralized, and has measurable heterogeneity on both sides. That combination makes it possible to observe whether booms change the quality of capital reallocation, not only the quantity.

2.2 Contract data

- Contract data come from IHS and Rigzone.
- The data identify new contracts, renegotiations or extensions, contract duration, day rates, rig identities, and whether rigs are active or available.
- The sample has 1,733 new contracts and 922 renegotiations/extensions.
- The mean day rate for new contracts is about \$62,000 per day; for renegotiations it is about \$52,000 per day.
- Average monthly utilization is about 76% across rig types and months.

Interpretation: Prices and durations matter because the model uses observed contracts to infer the value of matches and the outside option of capital owners.

2.3 Well data and project heterogeneity

- Well data come from the Bureau of Safety and Environmental Enforcement.
- Well characteristics include water depth, mud weight, location, geological information, and production-related measures.
- The paper measures well value using lease bid data, following the idea that bids contain information about expected oil and gas deposits.
- Well complexity is measured using an engineering **mechanical risk index**; higher values correspond to more difficult drilling projects.

Interpretation: Complexity is not a productivity residual. It is an engineering measure tied to drilling difficulty. This makes it a credible ranking of the project side of the market.

2.4 Rig heterogeneity

- Rigs are ranked by maximum water depth, which is a proxy for capability, technology, and efficiency.
- The structural estimation groups rigs into three types:
 - low-efficiency rigs: maximum depth up to about 200 feet;
 - mid-efficiency rigs: maximum depth between about 200 and 300 feet;
 - high-efficiency rigs: maximum depth at least about 300 feet.
- More capable rigs are better suited to complex wells, but they may be overbuilt and costly for simple wells.

Interpretation: The ranking allows the paper to study assortative matching: whether better capital is used on more complex projects.

2.5 Booms and busts

- Booms and busts are driven by natural gas price movements.
- Gas prices are treated as an exogenous demand shifter for drilling activity.
- Boom periods are months with gas prices above the sample average; bust periods are months with gas prices below the sample average.

Interpretation: This cycle shifts the attractiveness of drilling and the availability of future matches. The main question is whether this changes match quality.

3 Model and empirical specifications

3.1 Timing of the model

Each month has four main steps:

1. existing matches may be extended;
2. potential wells choose whether to enter;
3. entered wells direct search toward rig submarkets;
4. meetings occur and agents decide whether to accept a match.

Interpretation: The timing highlights the key dynamic trade-off. A rig that accepts a current match cannot search again during the contract, so accepting a bad match has an opportunity cost.

3.2 Value of a match

For a well of type x , a rig of type y , and contract period k after date t , the per-period match value is

$$v_{xyt,k} = m_{0,y} + m_{1,y}x_{\text{complexity}} + m_2 E_t[g_{t+k}]x_{\text{quantity}}. \quad (2)$$

- $m_{0,y}$ captures rig-type baseline value for simple wells.
- $m_{1,y}$ captures complementarities between rig type and well complexity.
- $m_2 E_t[g_{t+k}]x_{\text{quantity}}$ captures the expected value of oil and gas output.

Interpretation: Sorting is valuable when $m_{1,\text{high}}$ is high and $m_{1,\text{low}}$ is low or negative. In that case, complex wells benefit from high-efficiency rigs, while simple wells do not need high-spec rigs.

3.3 Project owner's value of matching

For a contract of duration τ , the well owner's value of matching is

$$W_{xyt} = \sum_{k=0}^{\tau-1} \beta^k (v_{xyt,k} - p_{xyt}) + \beta^\tau E_t[h_{xy,t+\tau} W_{xy,t+\tau}]. \quad (3)$$

Interpretation: The project owner values the current net surplus from the contract plus the option that the relationship is extended later.

3.4 Partially directed search

A project chooses which rig type to search for. Its perceived value from targeting a specific rig is

$$\hat{p}_{x_i y_j t} = p_{x_i y_j t} - g_1 \mathbf{1}\{x_i \notin A_{y_j t}\} + \varepsilon_{ijt}^{\text{target}}. \quad (4)$$

Aggregating across rigs of the same type gives targeting probability

$$q_{xyt} = \frac{n_{yt} \exp(g_0[p_{xyt} - g_1 \mathbf{1}\{x \notin A_{yt}\}])}{\sum_{k \in Y} n_{kt} \exp(g_0[p_{xkt} - g_1 \mathbf{1}\{x \notin A_{kt}\}])}. \quad (5)$$

- $g_0 = 0$ gives random search.
- $g_0 \rightarrow \infty$ gives perfectly directed search.
- g_1 lets agents avoid submarkets where a match is likely to be rejected.

Interpretation: The targeting parameters measure how well the decentralized market routes projects toward suitable capital. A high g_0 would mean a highly directed market; the estimates show a market closer to random search than to perfect direction.

3.5 Entry of projects

A potential project enters with probability

$$e_{xt} = \frac{\exp(\sum_{k \in Y} q_{xkt} p_{xkt} - c)}{1 + \exp(\sum_{k \in Y} q_{xkt} p_{xkt} - c)}. \quad (6)$$

Interpretation: Higher expected search value induces more wells to enter. The entry margin matters for counterfactuals because policies can alter the number and type of wells searching for rigs.

3.6 Capital owner's value of matching and searching

A rig owner's value from matching is

$$V_{xyt} = \sum_{k=0}^{\tau-1} \beta^k p_{xyt} + \beta^\tau E_t [h_{xy,t+\tau} V_{xy,t+\tau} + (1 - h_{xy,t+\tau}) U_{y,t+\tau}]. \quad (8)$$

The value of searching for a type- y rig is

$$U_{yt} = \int \max\{V_{zyt}, \beta E_t U_{y,t+1}\} h_{zyt} dz + h_{\emptyset yt} \beta E_t U_{y,t+1}. \quad (9)$$

Interpretation: The sorting effect operates through U_{yt} . In booms, the value of waiting rises because future matching opportunities are better. This raises the rejection threshold for bad matches.

3.7 Nash bargaining and surplus

Prices solve

$$p_{xyt} = \arg \max_p [V_{xyt} - \beta E_t U_{y,t+1}]^\delta W_{xyt}^{1-\delta}. \quad (10)$$

Total match surplus is

$$S_{xyt} = W_{xyt} + V_{xyt} - \beta E_t U_{y,t+1}. \quad (1)$$

The acceptance set is

$$A_{yt} = \{x : S_{xyt} \geq 0\}. \quad (2)$$

Interpretation: Prices are informative because they reflect both current match value and the rig owner’s outside option. This is why prices help separate true match quality from demand composition.

3.8 State vector and meeting technology

Agents summarize the market state as

$$s_t = [g_t, n_{\text{low},t}, n_{\text{mid},t}, n_{\text{high},t}], \quad (11)$$

where g_t is the gas price and n_{yt} is the number of available rigs of type y .

The meeting technology is approximated by

$$q_y^{\text{capital}}(v_{yt}) = \min \{1 - \exp(-a_y/v_{yt}), 1/v_{yt}\}, \quad (12)$$

$$q_y^{\text{project}}(v_{yt}) = \min \{v_{yt}[1 - \exp(-a_y/v_{yt})], 1\}. \quad (13)$$

Interpretation: The meeting technology captures search frictions. Even if a project targets a rig type, matching is probabilistic and depends on market tightness.

3.9 Welfare decomposition

The counterfactual welfare measure is total value of projects drilled minus entry costs and operating expenses. Counterfactual changes are decomposed into:

- **quality effect:** change in match values holding the number of matches fixed;
- **quantity effect:** value of additional or lost matches;
- **entry cost saving:** change in total entry costs.

Interpretation: This decomposition separates better allocation from more activity. That distinction is central because booms may improve welfare by raising match quality even if they do not simply increase output.

4 Identification and empirical design

4.1 Main identification problem

- Observed boom-time matching can improve because agents sort better.
- But it can also improve because the composition of wells entering the market changes.
- Since the full set of searching wells is not observed, raw matching patterns alone cannot distinguish sorting from demand composition.

Interpretation: The paper’s central empirical challenge is a missing-market problem: the econometrician sees matches but not all attempted searches or rejected projects.

4.2 How the model separates sorting from demand

- Match values are disciplined by price data and Nash bargaining.
- The rig owner’s outside option is recovered from observed contracts, extensions, prices, and state transitions.

- Acceptance sets can then be constructed from match surplus.
- Targeting parameters are identified from boom-bust sorting moments after controlling for acceptance sets.
- Demand-distribution parameters are identified from the distribution of observed matches and variation in utilization.

Interpretation: Prices are essential. Without prices, the model could confuse high demand for complex wells with high match value from pairing complex wells with high-efficiency rigs.

4.3 Role of natural gas prices

- Natural gas prices create cyclical variation in drilling demand.
- They raise the value of current drilling opportunities and the expected value of future opportunities.
- The paper uses these cycles to study whether agents become more selective in booms.

Interpretation: The empirical test is whether booms rotate matching patterns toward stronger sorting rather than just shifting the volume of drilling.

4.4 Limits of the design

- The paper is structural, so welfare results depend on the model's search, bargaining, and demand assumptions.
- The set of potential wells is not directly observed.
- The analysis focuses on shallow-water Gulf of Mexico drilling before 2010, so deep-water drilling or other capital markets may differ.
- The intermediary counterfactual abstracts from the private incentives and fixed costs of creating such a platform.

Interpretation: The paper is strongest as a quantitative mechanism study: it shows that sorting can be important and measures its welfare consequences in a rich market, rather than claiming a model-free causal experiment.

5 Tables and figures

This section groups adjacent visuals to save space. Each visual is directly cropped from the original paper.

5.1 Figure 1 and Table 1: sorting intuition and data scale

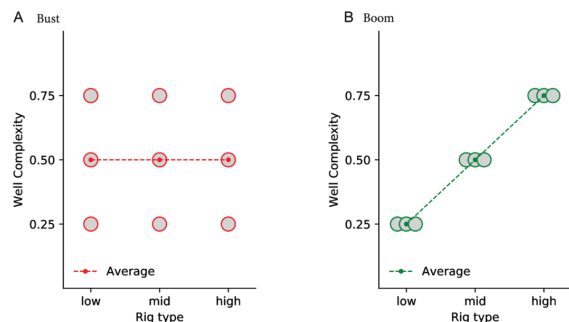


FIG. 1.—Illustration of the sorting effect. Suppose that in both panels there are three rigs of each type, [low, mid, high], and three wells of each type, [0.25, 0.5, 0.75], where a higher number corresponds to a more complex well. Each panel plots an allocation of the nine wells to the nine rigs. In a bust, all rigs drill similar wells, resulting in a flat average match line. In a boom, simple wells are allocated to low-efficiency rigs, and complex wells are allocated to high-efficiency rigs, resulting in a more diagonal average match line. For a fixed number of rigs and wells, as long as the match value is supermodular in rig type, there will be higher total output in the boom allocation.

(a) Figure 1: illustration of the sorting effect

TABLE 1
SUMMARY STATISTICS FOR DATASET

Variable	Units	N	Mean	SD	10%	90%
Rig price: new contracts	Thousands of USD per day	1,733	62	35	27	111
Duration: new contracts	Days	1,733	65	76	27	120
Rig price: renegotiations	Thousands of USD per day	922	52	26	28	81
Duration: renegotiations	Days	922	68	59	29	126
Value	Millions of USD	2,655	8.8	25	25	8
Complexity	Index	2,655	.87	.51	.34	1.46
Water depth	Feet	2,655	117	80	37	233
Monthly utilization	% rigs under contract	360	.76	.2	.49	1.0

NOTE.—Monthly utilization is for each of the three types of rig over 120 months.

(b) Table 1: summary statistics

Read:

- Figure 1 shows the core mechanism: when the market becomes thick, agents can reject poor matches and wait for better ones.
- The sorting effect reallocates high-quality rigs toward complex wells and low-quality rigs toward simple wells.
- Table 1 shows that the market has substantial variation in prices, duration, project value, complexity, and utilization.
- New contract day rates average about \$62,000 per day, while renegotiation day rates average about \$52,000 per day.
- Well values and well complexity are highly dispersed, which gives the market enough heterogeneity for matching quality to matter.

Economic message: The figure gives the economic intuition and the table shows why the mechanism is empirically relevant. The market is not a homogeneous commodity market: rigs and wells differ enough that who matches with whom can change welfare.

Detailed interpretation: The important point in Figure 1 is not simply that booms create more matches. The relevant change is a reordering of matches. If the market in a boom allows better parties to find one another, then productivity rises even without changing the technology of drilling. Table 1 shows that day rates and utilization are large enough for outside options to matter: a rig owner who rejects a contract is giving up tens of thousands of dollars per day, so the fact that agents still become more selective implies a substantial future option value.

5.2 Figures 2 and 3: positive assortative matching and cyclical prices

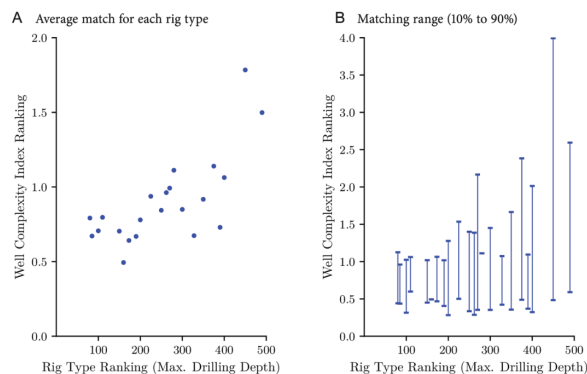


FIG. 2.—Positive assortative matching. Higher-ranked rigs match with more complex wells. Rigs are constructed in discrete increments of maximum drilling depth, so each point on the x-axis might correspond to many unique rigs.

(a) Figure 2: positive assortative matching

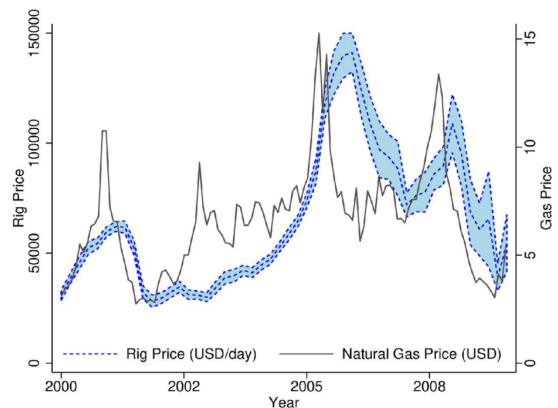


FIG. 3.—Booms and busts. How fluctuations in the natural gas price affect rig price industry.

(b) Figure 3: gas prices and rig prices

Read:

- Figure 2 shows that higher-ranked rigs drill more complex wells on average.
- The relationship is positive but not perfectly segmented: high-spec rigs still drill some simple wells, and lower-spec rigs still drill some more difficult wells.
- Figure 3 shows that rig prices move closely with natural gas prices.
- Rig prices rise above \$100,000 per day in high gas-price periods and fall toward roughly \$30,000 per day in low gas-price periods.

Economic message: Figure 2 establishes that matching quality is an economic object in the data. Figure 3 establishes the cyclical shock that changes market thickness and outside options.

Detailed interpretation: The incompleteness of sorting in Figure 2 is important. If sorting were already perfect, search frictions would be irrelevant and counterfactual improvements would be small. If there were no sorting at all, heterogeneous match quality might not matter. The data are in the middle: positive sorting exists, but the market leaves room for misallocation. Figure 3 then shows why a dynamic model is needed: a rig owner's acceptance decision depends on whether current prices are temporarily high and on whether future prices are expected to remain high.

5.3 Figure 4 and Tables 2–3: booms, busts, price dispersion, and mismatch

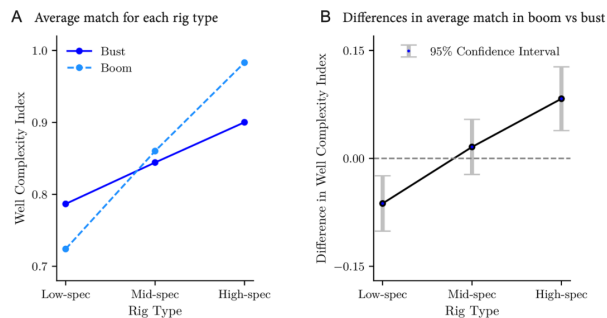


FIG. 4.—Matching patterns in booms and busts. *A*, When a bust turns to a boom, the market features stronger sorting patterns. *B*, Differences in matching patterns are statistically significant, using confidence intervals from table A-4. Figure A-3 (figs. A-1–A-6 are in the online appendixes) shows that the overall composition of matches in booms and busts is relatively similar.

Figure 4: matching patterns in booms and busts

TABLE 2
EVIDENCE OF PRICE DISPERSION

	Using Aggregated Rig Types (1)	Using Disaggregated Rig Types (2)
$1 - R^2$	.37	.27
$SD(\hat{p}_u)$	11	9
$SD(\hat{p}_u)$	18	18

NOTE.—Standard deviations are measured in thousands of US dollars per day. The dependent variable $\hat{p}_u$ is prices de-measured by the average price in each month. The residual from the price regression is denoted by $\hat{p}_u$.

(a) Table 2: evidence of price dispersion

TABLE 3
DOCUMENTING MISMATCH

	Change in Match Value (millions of USD)		
	Bust	Boom	Difference: Bust versus Boom
Optimal match versus empirical match	.779	.59	.188
<i>t</i> -test	.001***	.006***	.019**

NOTE.—Preview of full model-based results through documenting instances of mismatch by reallocating empirical matches to optimal matches. I measure the degree of mismatch using the estimated match values from the model. I report the average change at the contract level (for comparison, the average payment to a rig for a new contract is around \$3 million). I split the results into the improvements to matching in the bust, the boom, and the difference in the boom-bust change. I also report *p*-values from a *t*-test for the difference in mean match values.

** $p < .05$.

*** $p < .01$.

(b) Table 3: documenting mismatch

Read:

- Figure 4 shows that the average matching line rotates in booms: low-efficiency rigs move toward simpler wells, and high-efficiency rigs move toward more complex wells.
- This is the raw empirical signature of stronger sorting.
- Table 2 shows that even after using rig types and other match information, substantial price dispersion remains: the unexplained share of price variation is about 0.37 with aggregated rig types and 0.27 with disaggregated rig types.
- Table 3 shows that reallocating matches toward an optimal benchmark raises match value by about \$0.779 million in busts and \$0.590 million in booms; the difference is about \$0.188 million.

Economic message: The boom-bust rotation is the key reduced-form fact. Tables 2 and 3 motivate the structural model: prices contain information about match values, and observed allocations are not perfectly efficient.

Detailed interpretation: A naive reading of Figure 4 would be that booms simply cause better matching. The paper is more careful: the observed rotation could also reflect a different set of wells entering in booms. Table 2 matters because price dispersion means the contract data can help identify unobserved match values. Table 3 is an early welfare diagnostic: there is measurable mismatch, and the gains from improved allocation

are economically meaningful. The fact that mismatch gains are not identical in booms and busts foreshadows the structural result that the cycle changes both match values and outside options.

5.4 Figure 5 and Table 4: model structure

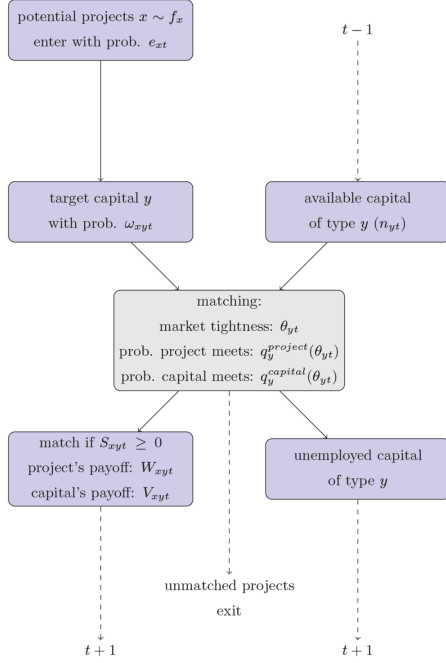


FIG. 5.—Search and matching within each submarket. At the beginning of each period, there is a distribution of searching projects and available capital n_y . The searching projects first choose which type of capital to target. Meetings are determined randomly within each submarket and are dependent on the market tightness θ_y . Finally, agents choose whether to match based on whether the total surplus of a match is positive ($S_y \geq 0$).

(a) Figure 5: search and matching within submarkets

TABLE 4
OVERVIEW OF HOW MODEL COMPONENTS ARE COMPUTED

Object	Parameters	Method
Discount rate, monthly	β	Calibrated: preliminary
Entry cost	c	Calibrated: preliminary
State transition beliefs	R_0, R_1, σ_c	Estimated: step 1
Bargaining weight	δ	Calibrated: step 1
Demand distribution	$\tau_2, \tau_3, \tau_4, \mu, \sigma, \rho_0, \rho_1, \rho_2, \rho_3$	Estimated: step 2
Demand draws	k_0, k_1	Estimated: step 2
Match value	$\{m_{0,y}, m_{1,y}, m_2\}$	Estimated: step 2
Extension parameter	η	Estimated: step 2
Targeting parameters	γ_0, γ_1	Estimated: step 2
Meeting technology	$\{a_{j,y}\}$	Estimated: step 2

NOTE.—Overview of the parameters to be estimated or calibrated.

(b) Table 4: model components

Read:

- Figure 5 describes the matching process inside each capital submarket: projects choose a rig type, meetings are probabilistic, and agents accept or reject based on match surplus.
- Table 4 summarizes the model's estimated and calibrated components: state transitions, bargaining, demand distribution, match values, extension probabilities, targeting, and meeting technology.
- The model treats search frictions and acceptance decisions separately; this is crucial because both channels affect sorting.

Economic message: The model is designed to make the unobserved search process observable through its implications for contracts, prices, utilization, and matching patterns.

Detailed interpretation: Figure 5 clarifies that matching is not only a demand-side choice. Projects may target a rig type, but meetings are not guaranteed, and rigs can reject the match. This creates two layers of friction: *contact frictions* and *acceptance frictions*. Table 4 is useful because it shows which objects come directly from data and which require structural estimation. The welfare results depend on all parts of this system, but the central mechanism is still transparent: higher U_{yt} in booms changes acceptance and targeting thresholds.

5.5 Figure 6 and Table 5: estimated model and search values

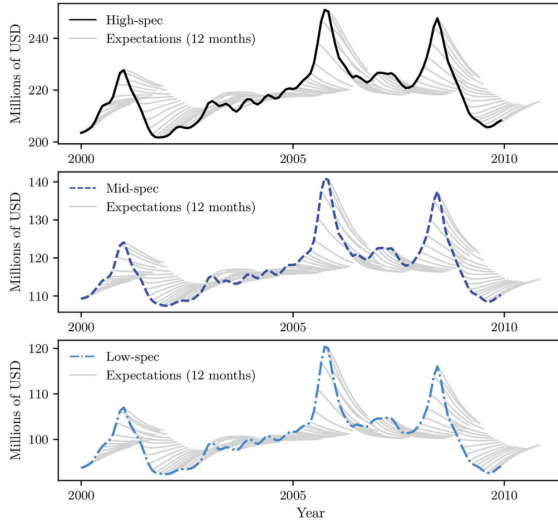


FIG. 6.—Rig's value of searching U_s . Points on the graph are plotted monthly. The gray lines correspond to the 12-month future beliefs of the value of searching.

(a) Figure 6: rig value of searching

TABLE 5
ESTIMATED PARAMETERS

Variable	Symbol	Value
Match value, extension	$m_{0,low}$	88.4 (74.2 to 98.8)
	$m_{0,mid}$	72.6 (56.8 to 88.9)
	$m_{0,high}$	57.2 (47.4 to 67.9)
	m_2	3.7 (.7 to 10.1)
	$m_{1,low}$	-37.5 (-45.7 to -29.3)
	$m_{1,mid}$	.7 (.4 to 1.2)
	$m_{1,high}$	40.5 (25.0 to 56.6)
	η	.38 (.35 to .4)
	μ	.68 (.52 to .77)
	σ	.72 (.57 to .95)
	ρ_0	.031 (.015 to .047)
	ρ_2	-.024 (-.044 to -.004)
	τ_2	.69 (.65 to .73)
	τ_5	.2 (.17 to .23)
	ρ_1	.037 (.001 to .073)
Demand	ρ_3	.0028 (.0003 to .0053)
	Entry cost, bargaining parameter	c \$1.32 million
	δ	NA .37
Potential project draws	k_0	NA 22.9 (11.1 to 31.0)
	k_1	8.0 (6.1 to 11.3)
Meeting technology	a_{low}	.44 (.31 to .62)
	a_{mid}	.57 (.41 to .71)
	a_{high}	16.5 (8.02 to 25.12)
Targeting parameter	γ_0	.71 (.26 to 1.11)
	γ_1	1.43 (.51 to 2.25)

NOTE.—All parameters are estimated using the simulated method of moments, except for estimates for ρ_0 , ρ_1 , ρ_2 , and ρ_3 , which are computed using the ordinary least squares (OLS) regression of $x_{quantity} = \rho_0 + \rho_1 x_{complexity} + \rho_2 x_{complexity}^2 + \rho_3 x_{complexity}^3$. Confidence intervals (95%) are shown in parentheses. An NA term in the confidence interval denotes a calibrated value. The confidence intervals are computed using 200 bootstrap replications, except for estimates for ρ_0 , ρ_1 , ρ_2 , and ρ_3 , which are computed using the standard errors from the OLS regression.

(b) Table 5: estimated parameters

Read:

- Figure 6 shows that rigs' values of searching rise in booms and fall in busts.
- The gray lines show forward-looking expected search values, indicating mean reversion in expectations.
- Table 5 estimates that low-spec rigs are valuable for simple wells, while high-spec rigs have a large positive slope with well complexity.
- The estimated extension probability is about 0.38.
- The estimated targeting parameter g_0 is about 0.71, closer to random search than perfect directed search.
- The high-rig meeting technology parameter is much larger than the low- and mid-rig parameters, implying less severe contact frictions in the high-spec submarket.

Economic message: Figure 6 is the direct model-based evidence for the option-value channel. Table 5 shows why sorting is valuable: high-efficiency rigs are much better suited to complex wells, while low-efficiency rigs are better matched to simple wells.

Detailed interpretation: The critical visual in Figure 6 is the procyclicality of U_{yt} . In a boom, the cost of taking a bad match is high because the rig may miss a better future contract. That raises selectivity and pushes the market toward stronger sorting. Table 5 gives the economic shape of match values: high-spec rigs are not universally best; they are best for complex wells. This is why an intermediary or improved search technology can generate large quality gains without necessarily increasing match counts by much.

5.6 Figures 7–8: sorting and intermediary counterfactuals

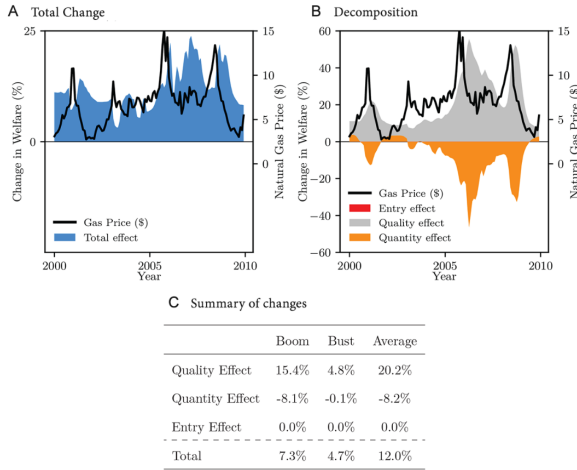


FIG. 7.—No-sorting counterfactual results. A, Total change. B, Decomposition. Shown is the change in welfare when moving from the no-sorting counterfactual to the market baseline. I keep the composition of searching wells fixed to the market benchmark, resulting in an entry effect of 0. The welfare in dollars at the market baseline is 5.0 billion.

(a) Figure 7: no-sorting counterfactual

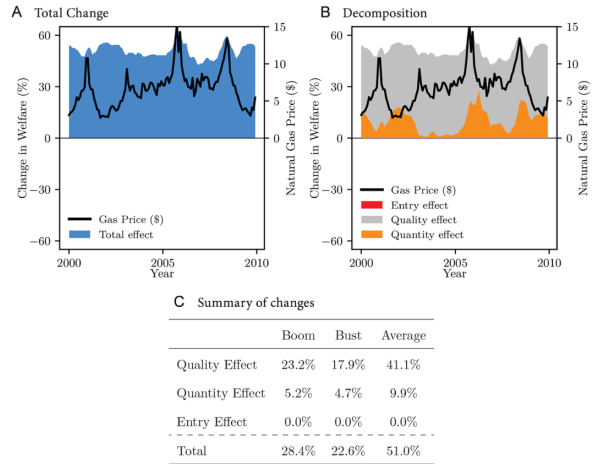


FIG. 8.—Intermediary counterfactual results. A, Total change. B, Decomposition. Shown is the change in welfare when moving from the market baseline to the intermediary counterfactual. I keep the composition of searching wells fixed to the market benchmark, resulting in an entry effect = 0. The welfare in dollars at the market baseline is 5.0 billion.

(b) Figure 8: intermediary counterfactual

Read:

- Figure 7 compares the market allocation with a counterfactual that shuts down selective sorting.
- The sorting effect raises welfare by about 12.0% overall, with a larger contribution in booms than in busts.
- The decomposition shows fewer matches but higher match quality; match quality dominates the lost quantity.
- Figure 8 shows that an intermediary that reduces search frictions raises welfare by about 51.0%.
- The intermediary gain is mostly a quality gain: match quality improves by about 41.1%, while the quantity effect is about 9.9%.

Economic message: Sorting is already valuable in the decentralized market, but search frictions still leave large gains from better market design.

Detailed interpretation: Figure 7 illustrates a central trade-off: selectivity can reduce the number of matches, but this is not necessarily bad. In this market, rejecting low-quality matches improves the average match enough to raise total welfare. Figure 8 pushes the logic further. If a platform or intermediary could reduce search frictions, the welfare gain would come primarily from better matches, not simply more activity.

This is a market-design result: the bottleneck is not just demand, but information and routing across heterogeneous capital and heterogeneous projects.

5.7 Figure 9: demand-smoothing policy

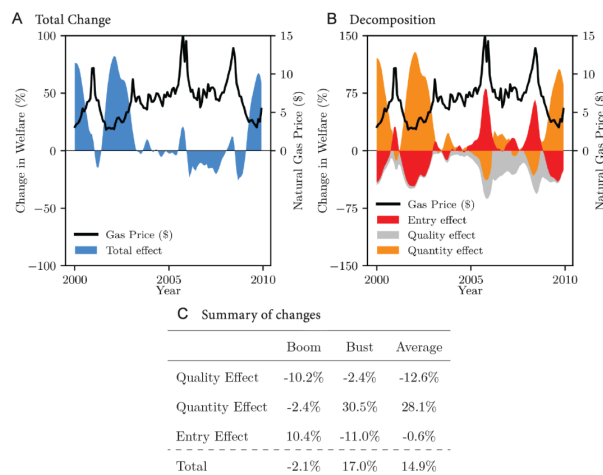


FIG. 9.—Demand-smoothing counterfactual results. *A*, Total change. *B*, Decomposition. Shown is the change in welfare when moving from the market baseline to the demand-smoothing counterfactual. The welfare in dollars at the market baseline is 5.0 billion. The entry effect corresponds to the total change in entry costs, so it will be negative when there is more entry.

Figure 9: demand-smoothing counterfactual

Read:

- Figure 9 studies a counterfactual policy that holds the natural gas price at its long-run average.
- The policy changes entry and matching quantities substantially.
- The overall welfare gain is modest, about 14.9%.
- The quality effect is negative in booms because lower boom prices reduce selectivity and worsen matching quality.
- Match quality also falls slightly in busts because smoothing changes the composition of entering projects.

Economic message: Demand smoothing is not a clean substitute for improving matching. It changes activity, but it can weaken the boom-time sorting channel that raises productivity.

Detailed interpretation: Figure 9 is especially important for policy. A demand-smoothing policy seems attractive because booms and busts appear disruptive. But the model shows that the cycle itself generates useful information and useful option values. When policy suppresses booms, it can also suppress selectivity. The result is not that demand smoothing is always bad; rather, the result is that smoothing should be evaluated through the matching mechanism, not only through aggregate output volatility.

6 Short summary

- The paper studies how decentralized capital markets reallocate heterogeneous capital during booms and busts.

- The setting is the market for offshore drilling rigs in the U.S. Gulf of Mexico from 2000–2009.
- Rigs are ranked by capability, and wells are ranked by engineering complexity.
- The data show positive assortative matching: better rigs drill more complex wells.
- Booms, measured by high natural gas prices, are associated with stronger sorting.
- The mechanism is dynamic: booms raise the value of waiting for a better match, so agents reject bad matches more often.
- The structural model includes entry, partially directed search, probabilistic meetings, acceptance sets, contract extensions, and Nash bargaining.
- Prices and state transitions are used to infer match values and rig-owner outside options.
- Estimated match values show that high-efficiency rigs are particularly valuable for complex wells, while low-efficiency rigs are better suited to simple wells.
- Removing sorting reduces welfare; the market sorting effect raises welfare by about 12%.
- A market intermediary that improves search and matching could raise welfare by about 51%, mostly through better match quality.
- Demand smoothing changes activity but yields only modest welfare gains and can reduce match quality in booms.

7 Conclusion

7.1 Core conclusion

Vreugdenhil shows that booms and busts affect not only the amount of capital reallocation but also its quality. In the offshore drilling market, booms make future opportunities more valuable. That raises agents' option value of search, makes them more selective, and pushes the market toward stronger positive assortative matching. This sorting effect is quantitatively important: it raises welfare relative to a no-sorting counterfactual.

7.2 Economic intuition

The key intuition is that capital owners face an intertemporal acceptance problem. A rig can accept a current contract, or it can wait for another well. In a bust, the value of waiting is low, so agents accept a broader set of matches. In a boom, the value of waiting rises, so agents reject weak matches and hold out for better matches. When high-efficiency rigs are especially productive on complex wells, this selectivity improves allocation and raises total surplus.

7.3 What the paper teaches about booms and busts

A standard view is that booms create congestion and busts create idle resources. This paper adds a more subtle force: booms can improve sorting by making the market thicker and raising outside options. Cycles may therefore have a productivity component through matching quality. The implication is that aggregate productivity movements can reflect not only technology or entry, but also how well heterogeneous capital is matched to heterogeneous projects.

7.4 Market-design implication

The intermediary counterfactual is one of the most important lessons. A platform that improves search, routing, or information about suitable matches could create welfare gains much larger than the sorting effect

already present in the decentralized market. The gain is mostly a match-quality gain, which means the relevant policy or business innovation is not simply to increase activity. It is to help the right rig find the right well faster.

7.5 Policy implication

Demand smoothing is a blunt policy. It can reduce volatility in activity, but it may also weaken the mechanism that improves matching in booms. If the main problem is poor search and misallocation, then policies that improve information, matching technology, contract standardization, or platform access may be more effective than policies that smooth demand. The paper therefore cautions against evaluating countercyclical subsidies or price smoothing purely by their effects on drilling volume.

7.6 Implications for capital-market efficiency

The results imply that decentralized capital markets can be partly efficient even with frictions. Search frictions do not eliminate sorting; agents adapt by becoming selective when the outside option is valuable. However, the intermediary counterfactual shows that decentralized matching remains far from the efficient frontier. Market participants do internalize their own match surplus, but they do not necessarily create the information and coordination infrastructure needed to maximize system-wide allocation quality.

7.7 External validity

The mechanism is likely relevant beyond offshore drilling whenever production requires matching heterogeneous projects to heterogeneous capital: shipping, aircraft leasing, specialized equipment rental, construction machinery, venture capital, managerial talent, and some labor markets. The exact magnitude will depend on whether capital is reusable, whether search is decentralized, whether projects are vertically heterogeneous, and whether match quality is observable through prices.

7.8 Caveats and limitations

The paper uses a rich structural model, so counterfactuals depend on assumptions about search, bargaining, beliefs, and demand. The full set of searching wells is not directly observed. The analysis is limited to shallow-water Gulf of Mexico drilling from 2000–2009. The intermediary counterfactual also abstracts from the fixed costs, adoption frictions, and strategic incentives that might prevent a real platform from emerging.

7.9 Research agenda

The paper opens several research directions. First, future work could study why intermediaries do or do not emerge in decentralized capital markets. Second, researchers could examine whether platform technologies improve match quality in other capital markets. Third, policy studies could compare demand smoothing with matching-infrastructure policies. Fourth, macro models of productivity could incorporate boom-time sorting as a channel through which cycles change aggregate efficiency.

7.10 Takeaway

The paper’s central message is that reallocation is a matching problem. Booms and busts change not only how much capital moves, but also how carefully it is matched. In this market, the most valuable interventions are likely those that improve information and matching rather than those that simply smooth demand.